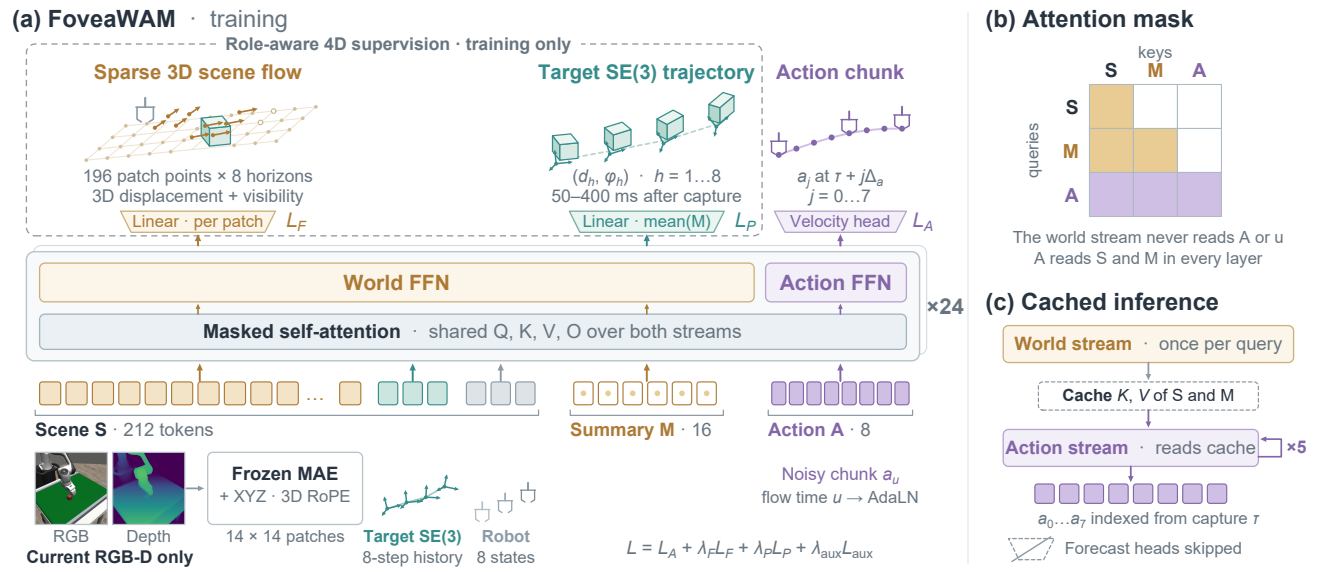

(a) FoveaWAM · training
Role-aware 4D supervision · training only
Sparse 3D scene flow
196 patch points × 8 horizons
3D displacement + visibility
Linear · per patch
L_F
Target SE(3) trajectory
(d_h, φ_h) · $h = 1…8$
50–400 ms after capture
Linear · mean(M)
L_P
Action chunk
a_j at $\tau + j\Delta_a$
$j = 0…7$
Velocity head
L_A
World FFN
Action FFN
Masked self-attention · shared Q, K, V, O over both streams
×24
Scene S · 212 tokens
Summary M · 16
Action A · 8
RGB
Depth
Current RGB-D only
Frozen MAE
+ XYZ · 3D RoPE
14 × 14 patches
Target SE(3)
8-step history
Robot
8 states
Noisy chunk a_u
flow time u → AdaLN
$L = L_A + \lambda_F L_F + \lambda_P L_P + \lambda_{\text{aux}} L_{\text{aux}}$
(b) Attention mask
keys
S
M
A
queries
S
M
A
The world stream never reads A or u
A reads S and M in every layer
(c) Cached inference
World stream · once per query
Cache K, V of S and M
Action stream · reads cache
×5
$a_0 … a_7$ indexed from capture τ
Forecast heads skipped